Assignment 2 - FMAN45

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May 2026

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We calculate the kernel $\mathbf{K} = [k(x_i, x_j)]_{1 \leq i,j \leq 4}$ by the definition in the instructions. Hence, $k(x_i, x_j) = \Phi(x_i)\Phi^T(x_j) = x_i x_j + x_i^2 x_j^2$. When using this definition for the given values in the instruction, we get the following kernel:

$$K = \begin{pmatrix} 20 & 6 & 2 & 12 \\ 6 & 2 & 0 & 2 \\ 2 & 0 & 2 & 6 \\ 12 & 2 & 6 & 20 \end{pmatrix} \tag{1}$$

2

Firstly, we observe that the constraint is satisfied since $\sum_{i=1}^{4} y_i \alpha_i = (1 - 1 - 1 + 1)\alpha = 0$. Here we use the fact that $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4$. We can now rewrite the first term in equation 5 in the instructions as:

$$\sum_{i=1}^{4} \alpha_i = 4\alpha \tag{2}$$

The second term is as follows:

$$\frac{1}{2} \sum_{i,j=1}^{4} \alpha_i \alpha_j y_i y_j k(x_i, x_j) = \frac{1}{2} \alpha^2 \sum_{i,j=1}^{4} y_i y_j K_{i,j} \tag{3}$$

We can now calculate the sum in equation 3 using the values for y_i in the instructions and the kernel matrix K calculated above. We do this first by fixing $i = 1$ and then loop over $j \in \{1, 2, 3, 4\}$. Then fix $i = 2$ and then loop over $j \in \{1, 2, 3, 4\}$ and so on. If we follow this algorithm, we end up with:

$$\frac{1}{2} \sum_{i,j=1}^{4} \alpha_i \alpha_j y_i y_j k(x_i, x_j) = 18\alpha^2 \tag{4}$$

We now want to maximize:

$$\begin{aligned} L &= 4\alpha - 18\alpha^2 \\ \frac{\partial L}{\partial a} &= 0 \rightarrow \\ 4 - 36\alpha &= 0 \rightarrow \\ \alpha &= \frac{1}{9} \end{aligned} \tag{5}$$

This value for α satisfies the constraints.

3

Firstly, we derive the right kernel and plug it in to equation 7 in the instruction:

$$g(x) = \frac{1}{9}\sum_{j=1}^{4} y_j(xx_j + x_j^2x^2) + b \tag{6}$$

Then we take the values from table 1 in the instructions and plug them into equation 6 for each j .

$$\begin{aligned} g(x) &= \frac{1}{9}(-2x + 4x^2 + x - x^2 - x - x^2 + 2x + 4x^2) + b \\ &= \frac{1}{9}6x^2 + b \end{aligned} \tag{7}$$

Since $\alpha \neq 0$ we have a support vector and can use equation 6 in the instruction to solve b. We use $x_i = 2$ and $y_1 = 1$.

$$\begin{aligned} g(-2) &= \frac{1}{9}6(-2)^2 + b = 1 \\ g(-2) &= \frac{8}{3} + b = 1 \rightarrow \\ b &= -\frac{5}{3} \end{aligned} \tag{8}$$

The result is then:

$$g(x) = \frac{2}{3}x^2 - \frac{5}{3} \tag{9}$$

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The new data set has the same kernel and structure as the previous one, with the difference of another negative classified point between the two previous ones. In addition, there are two added positive classified data points above the previous one. Therefore, we assume that $g(x)$ must have the same structure as the previous solution: $g(x) = ax^2 + bx + c$. However, for every x there is a $-x$ with the same label. Therefore, the classifier must satisfy $g(x) = g(-x)$. This

implies $b = 0$. If we pick the coordinates closest to the boundary, we get the support vectors and can use equation 6 in the instructions to find the constants:

$$\begin{aligned} &\text{for: } x = -2, y_i = 1 \\ &4a + c = 1 \\ &\text{for: } x = -1, y_i = -1 \\ &-a - c = 1 \end{aligned} \tag{10}$$

We solve the following system:

$$\begin{aligned} &4a + c = 1 \rightarrow c = 1 - 4a \\ &-a + 4a - 1 = 1 \rightarrow a = \frac{2}{3} \\ &c = \frac{3}{3} - \frac{8}{3} = \frac{-5}{3} \end{aligned} \tag{11}$$

We can write the classifier as:

$$g(x) = \frac{2}{3}x^2 - \frac{5}{3} \tag{12}$$

5

Firstly, we introduce the Lagrangian multipliers $\alpha_i \geq 0$ for constrain $y_i \left(\mathbf{w}^T \mathbf{x}_i + b\right) \geq 1 - \xi_i$ and $\lambda_i \geq 0$ for $\xi_i \geq 0$. We can then rewrite the constraints:

$$\begin{aligned} &0 \geq 1 - \xi_i - y_i \left(\mathbf{w}^T \mathbf{x}_i + b\right) \\ &0 \geq -\xi_i \end{aligned} \tag{13}$$

We can then write the Lagrangian as:

$$L(w, b, \xi, \alpha, \lambda) = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n} \xi_i + \sum_{i=1}^{n} \alpha_i (1 - \xi_i - y_i \left(\mathbf{w}^T \mathbf{x}_i + b\right)) - \sum_{i=1}^{n} \lambda_i \xi_i \tag{14}$$

We can now take the derivatives and set them zero:

$$\begin{aligned} &\frac{\partial L}{\partial \mathbf{w}} = \mathbf{w} - \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i = 0 \rightarrow \mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i \\ &\frac{\partial L}{\partial b} = -\sum_{i=1}^{n} \alpha_i y_i = 0 \rightarrow \sum_{i=1}^{n} \alpha_i y_i = 0 \\ &\frac{\partial L}{\partial \xi_i} = C - \alpha_i - \lambda_i = 0 \rightarrow \alpha_i + \lambda_i = C \\ &\text{And since } \lambda \geq 0 \text{ , } 0 \leq \alpha_i \leq C \end{aligned} \tag{15}$$

We can then write:

$$\frac{1}{2}\|\mathbf{w}\|^2 = \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \tag{16}$$

We can then rewrite equation 14 as:

$$\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j \mathbf{x}_i^T\mathbf{x}_j+\xi_i\sum_{i=1}^{n}C-\alpha_i-\lambda_i+\sum_{i=1}^{n}\alpha_i-b\sum_{i=1}^{n}\alpha_i y_i-\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j \mathbf{x}_i^T\mathbf{x}_j \tag{17}$$

And with the relations discovered in 15 this expression becomes:

$$L(\alpha)=\sum_{i}^{n}\alpha_i-\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j \mathbf{x}_i^T\mathbf{x}_j \tag{18}$$

We then want to maximize over α to find the solutions, under the constraints:

$$\begin{aligned}&\max_{a_1,\ldots,a_n}\sum_{i=1}^{n}\alpha_i-\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j \mathbf{x}_i^T\mathbf{x}_j\\&\text{subjected to } 0\leq\alpha_i\leq C,\\&\sum_{i=1}^{n}\alpha_i y_i=0\end{aligned} \tag{19}$$

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From the KKT conditions, we get, with the Lagrangian multipliers α_i, λ_i

$$\begin{aligned}&\alpha_i(y_i(\mathbf{w}^T\mathbf{x}_i+b)-1+\xi_i)=0\\&\lambda_i\xi_i=0\end{aligned} \tag{20}$$

Since $y_i(\mathbf{w}^T\mathbf{x}_i+b)<1$ and $y_i\left(\mathbf{w}^T\mathbf{x}_i+b\right)\geq 1-\xi_i$, then $\xi_i>0$. From the complementary slackness in equation 20, this implies that $\lambda_i=0$. From the stationary condition in equation 15 we know that $\alpha_i+\lambda_i=C$, and since $\lambda_i=0$ we get $\alpha_i=C$.

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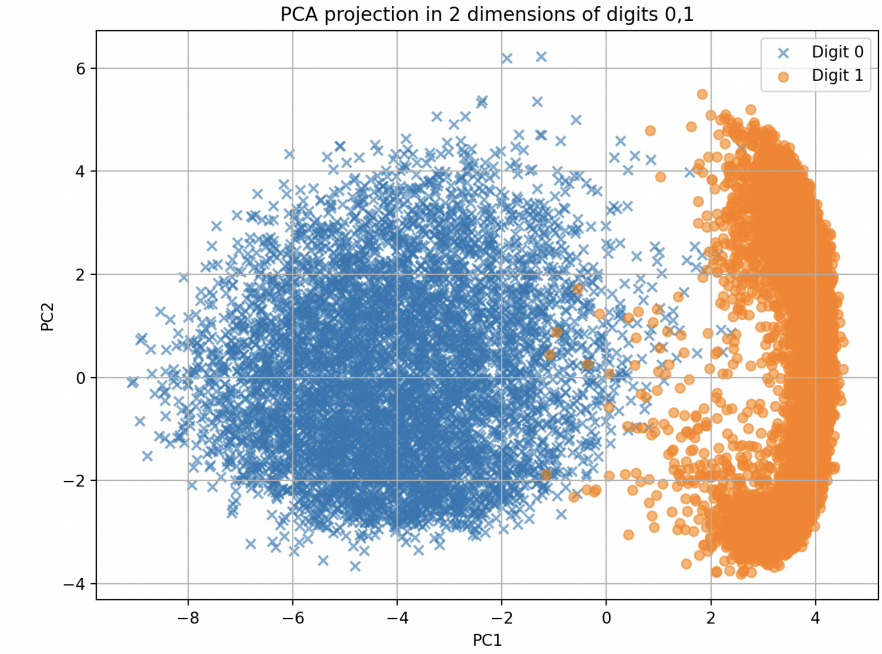


Figure 1: The PCA projection in 2 dimensions of digit 0 and 1 from the MNIST data set

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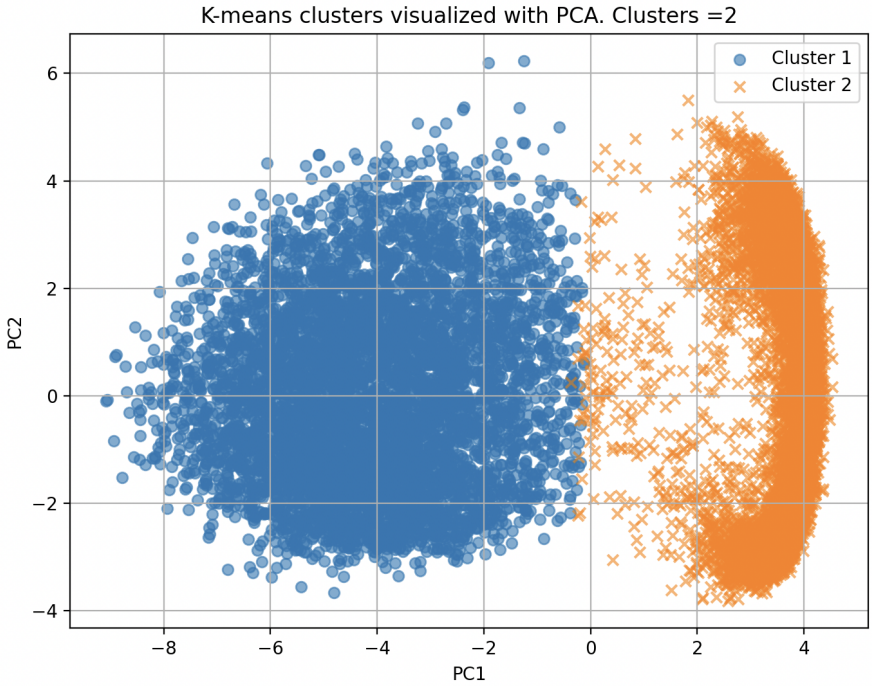


Figure 2: K-mean clustering with the data points from the PCA

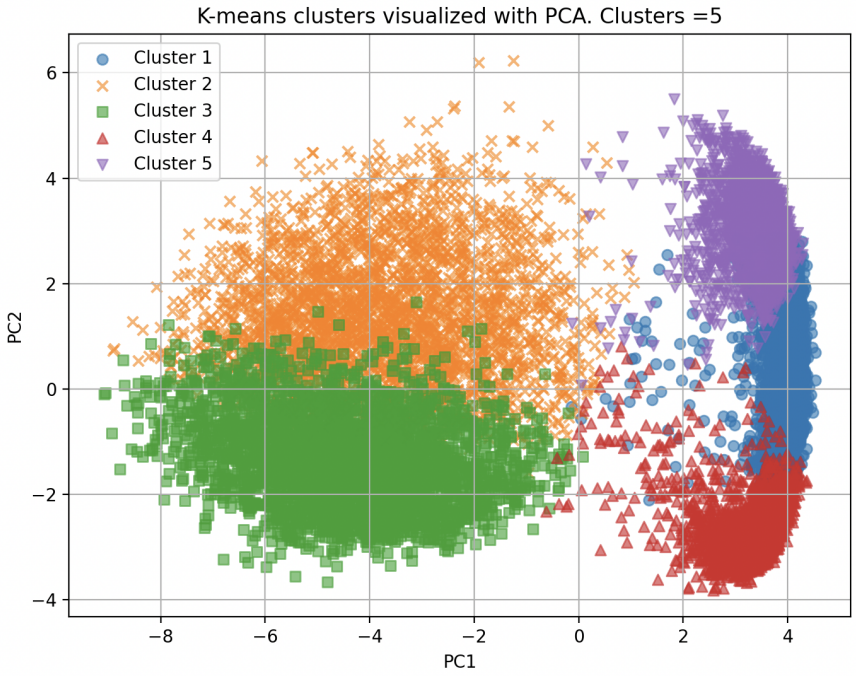


Figure 3: K-mean clustering with the data points from the PCA.

In Figure 3, the overlap between the clusters is clearly visible. The K-mean clustering does not overlap mathematically, but since we are reducing the dimension from 784 to 2, when plotting, they appear overlapping. In a higher dimension, they are not overlapping. Furthermore, PCA preserves variance in the dataset, it does not separate clusters.

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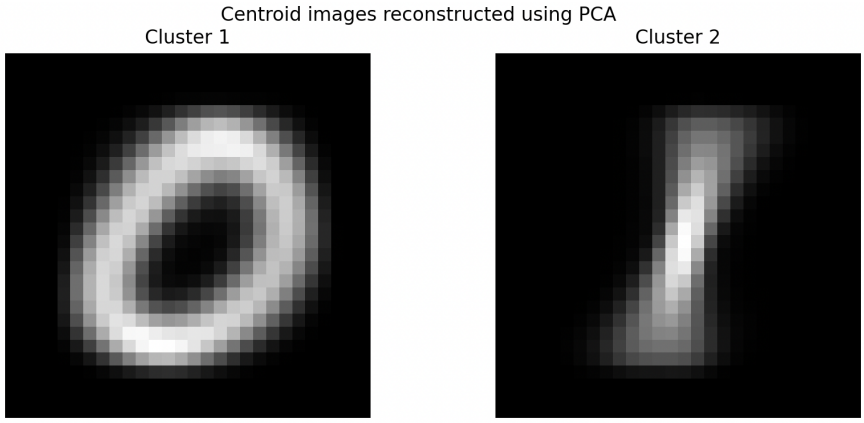


Figure 4: For two clusters we observe the different digits 0,1.

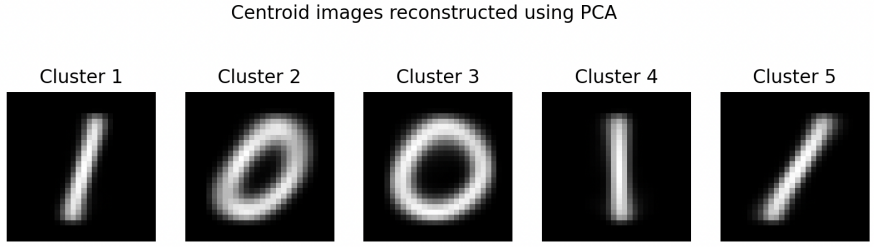


Figure 5: For five clusters we observe the different digits 0,1.

For Figure 5 we observe 3 pictures of digit one and 2 for digit zero. This is correlated with clusters 1,4 and 5 in Figure 3 and the zeros appear in clusters 2 and 3. Due to averaging, the images look clearer for 5 clusters than for 2.

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Table 1: K-means classification results

Training data	Cluster	# '0'	# '1'	Assigned to class	# misclassified
	1	112	6736	0	112
	2	5811	6	1	6
$N_{\text{train}} = 12665$				Sum misclassified:	118
				Misclassification rate (%):	0.931
Testing data	Cluster	# '0'	# '1'	Assigned to class	# misclassified
	1	12	1135	0	12
	2	968	0	1	0
$N_{\text{test}} = 2115$				Sum misclassified:	12
				Misclassification rate (%):	0.567

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number of clusters	Misclassification rate
2	0.567
4	0.330
6	0.189
8	0.189

The misclassification rate decreases with the number of clusters, but converges to 0.189.

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Table 2: Linear SVM classification results

Training data	Predicted class	True class: # '0'	# '1'
	'0'	5923	0
	'1'	0	6742
$N_{\text{train}} = 12665$		Sum misclassified:	0
		Misclassification rate (%):	0
Testing data	Predicted class	True class: # '0'	# '1'
	'0'	979	1
	'1'	1	1134
$N_{\text{test}} = 2115$		Sum misclassified:	2
		Misclassification rate (%):	0.095

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Gamma = 0.03 was used for Table 3

Table 3: Gaussian kernel SVM classification results

Training data	Predicted class	True class: # '0'	# '1'
	'0'	5923	0
	'1'	0	6742
$N_{\text{train}} = 12665$		Sum misclassified:	0
		Misclassification rate (%):	0
Testing data	Predicted class	True class: # '0'	# '1'
	'0'	980	0
	'1'	0	1135
$N_{\text{test}} = 2115$		Sum misclassified:	0
		Misclassification rate (%):	0

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The SVM learns a hyperplane in the feature space (decision boundary) based on the training data. This may be similar to the test data and more specifically it captures noise and characteristic in the train data. New images can have other characteristics and may not be separated by the decision boundary. The classifier may overfit the train data.